T3 group factors: EFT_1_after_F

Active theory: SM + S_1 + S_2 + $C_{LLS_1S_2}$. The fermion F is absent, so y_1, y_2 and M_F are not running EFT couplings.

$$j = \frac{d-1}{2}, \qquad C_2(d) = \frac{d^2-1}{4}, \qquad T(d) = \frac{d(d^2-1)}{12}.$$

In every comparison table, $\mathcal{S}$ denotes the RGE structure multiplying the displayed group factor in $16\pi^2\beta$.

Notation key

symbol	interaction / definition
g_1, g_2, g_3	SM gauge interactions
Y_u, Y_d, Y_e	SM Yukawa interactions
y_1	LFS_1 + h.c.
y_2	LFS_2 + h.c.
M_F	$-\bar{F}M_F F$
T	$\mathrm{Tr}(Y_e^\dagger Y_e + 3Y_u^\dagger Y_u + 3Y_d^\dagger Y_d)$
$T_\nu^{(1)}$	$\mathrm{Tr}(y_1^\dagger y_1)$
$T_\nu^{(2)}$	$\mathrm{Tr}(y_2^\dagger y_2)$
T_ν	$T_\nu^{(1)} + T_\nu^{(2)}$
m_1^2, m_2^2	$m_1^2 S_1^\dagger S_1,\ m_2^2 S_2^\dagger S_2$
λ_1	$\frac{1}{2}\lambda_1 (H^\dagger H)^2$
$\lambda_{S_i}^{(1)}$	$\frac{1}{2}\lambda_{S_i}^{(1)} (S_i^\dagger S_i)^2$
$\lambda_{HS_i}^{(1)}$	$(H^\dagger H)(S_i^\dagger S_i)$
$\lambda_{HS_i}^{(A)}$	$(H^\dagger T^A H)(S_i^\dagger T^A S_i)$
λ_5	$HHS_1S_2^\dagger$ + h.c.
$\lambda_{12}^{(1)}$	$(S_1^\dagger S_1)(S_2^\dagger S_2)$
$\lambda_{12}^{(A)}$	$(S_1^\dagger T^A S_1)(S_2^\dagger T^A S_2)$
$\lambda_{12}^{(\times)}$	$(S_1^\dagger S_2)(S_2^\dagger S_1)$

Stage field content

run	S_1	S_2	F	integrated at this stage	active after matching
T3-E	$(3,1)$	$(3,2)$	$(2,\frac{3}{2})$	F	S1, S2

Field representations

run	S_1	S_2	α
$\mathrm{T3-E}$	$(3,1)$	$(3,2)$	2

Gauge and Yukawa beta functions in this EFT

$\beta_{g_1}^{(1)}$

$\mathcal{S}$	$\mathrm{T3-E}$ $(\alpha = 2)$
g_1^3	$\frac{71}{6}$

$\beta_{g_2}^{(1)}$

$\mathcal{S}$	$\mathrm{T3-E}\ (\alpha=2)$
g_2^3	$-\frac{11}{6}$

$\beta_{g_3}^{(1)}$

$\mathcal{S}$	$\mathrm{T3-E}\ (\alpha=2)$
g_3^3	-7

$\beta_{Y_u}^{(1)}$

$\mathcal{S}$	$\mathrm{T3-E}\ (\alpha=2)$
$(Y_d(Y_d)^\dagger Y_u)_{ij}$	$-\frac{3}{2}$
$(Y_u(Y_u)^\dagger Y_u)_{ij}$	$\frac{3}{2}$
$g_2^2(Y_u)_{ij}$	$-\frac{9}{4}$
$g_3^2(Y_u)_{ij}$	-8
$g_1^2(Y_u)_{ij}$	$-\frac{17}{12}$
$T(Y_u)_{ij}$	1

$\beta_{Y_d}^{(1)}$

$\mathcal{S}$	$\mathrm{T3-E}\ (\alpha=2)$
$(Y_d(Y_d)^\dagger Y_d)_{ij}$	$\frac{3}{2}$
$(Y_u(Y_u)^\dagger Y_d)_{ij}$	$-\frac{3}{2}$
$g_2^2(Y_d)_{ij}$	$-\frac{9}{4}$
$g_3^2(Y_d)_{ij}$	-8
$g_1^2(Y_d)_{ij}$	$-\frac{5}{12}$
$T(Y_d)_{ij}$	1

$\beta_{Y_e}^{(1)}$

$\mathcal{S}$	$\mathrm{T3-E}\ (\alpha=2)$
$(Y_e(Y_e)^\dagger Y_e)_{ij}$	$\frac{3}{2}$
$g_2^2(Y_e)_{ij}$	$-\frac{9}{4}$
$g_1^2(Y_e)_{ij}$	$-\frac{15}{4}$
$T(Y_e)_{ij}$	1

$\beta_{m_1^2}$

$\mathcal{S}$	$\mathrm{T3-E}\ (\alpha=2)$
$\lambda_{S_1}^{(1)}\, m_1^2$	8
$g_2^2\, m_1^2$	-12
$g_1^2\, m_1^2$	-6
$\lambda_{12}^{(1)}\, m_2^2$	6
$\lambda_{S_1}^{(A)}\, m_1^2$	4

$\lambda_{12}^{(\times)}\, m_2^2$	4

$\beta_{m_2^2}$

$\mathcal{S}$	T3 − E ($\alpha = 2$)
$\lambda_{S_2}^{(1)}\, m_2^2$	8
$g_2^2\, m_2^2$	-12
$g_1^2\, m_2^2$	-24
$\lambda_{12}^{(1)}\, m_1^2$	6
$\lambda_{S_2}^{(A)}\, m_2^2$	4
$\lambda_{12}^{(\times)}\, m_1^2$	4

$\beta_{\lambda_{HS_1}^{(1)}}$

$\mathcal{S}$	T3 − E ($\alpha = 2$)
$\lvert\lambda_5\rvert^2$	4
$g_2^2\, \lambda_{HS_1}^{(1)}$	$-\frac{33}{2}$
$g_1^2\, \lambda_{HS_1}^{(1)}$	$-\frac{15}{2}$
$\lambda_1\, \lambda_{HS_1}^{(1)}$	6
$\left(\lambda_{HS_1}^{(1)}\right)^2$	4
$\lambda_{HS_1}^{(1)}\, \lambda_{S_1}^{(1)}$	8
$\lambda_{12}^{(1)}\, \lambda_{HS_2}^{(1)}$	6
g_2^4	6
g_1^4	3
$\left(\lambda_{HS_1}^{(A)}\right)^2$	2
$\lambda_{HS_1}^{(1)}\, \lambda_{S_1}^{(A)}$	4
$\lambda_{HS_2}^{(1)}\, \lambda_{12}^{(\times)}$	4
$\lambda_{HS_1}^{(1)}\, T$	2

$\beta_{\lambda_{HS_2}^{(1)}}$

$\mathcal{S}$	T3 − E ($\alpha = 2$)
$\lvert\lambda_5\rvert^2$	4
$g_2^2\, \lambda_{HS_2}^{(1)}$	$-\frac{33}{2}$
$g_1^2\, \lambda_{HS_2}^{(1)}$	$-\frac{51}{2}$
$\lambda_1\, \lambda_{HS_2}^{(1)}$	6
$\left(\lambda_{HS_2}^{(1)}\right)^2$	4
$\lambda_{HS_2}^{(1)}\, \lambda_{S_2}^{(1)}$	8
$\lambda_{12}^{(1)}\, \lambda_{HS_1}^{(1)}$	6
g_2^4	6
g_1^4	12
$\left(\lambda_{HS_2}^{(A)}\right)^2$	2
$\lambda_{HS_2}^{(1)}\, \lambda_{S_2}^{(A)}$	4
$\lambda_{HS_1}^{(1)}\, \lambda_{12}^{(\times)}$	4
$\lambda_{HS_2}^{(1)}\, T$	2

$\beta_{\lambda_{12}^{(1)}}$

$\mathcal{S}$	$\mathrm{T3-E}\ (\alpha=2)$
$\lvert\lambda_5\rvert^2$	$\frac{4}{3}$
$g_2^2\,\lambda_{12}^{(1)}$	-24
$g_1^2\,\lambda_{12}^{(1)}$	-30
$\left(\lambda_{12}^{(1)}\right)^2$	4
$\lambda_{12}^{(1)}\,\lambda_{S_1}^{(1)}$	8
$\lambda_{12}^{(1)}\,\lambda_{S_2}^{(1)}$	8
$\lambda_{HS_1}^{(1)}\,\lambda_{HS_2}^{(1)}$	4
g_2^4	12
g_1^4	48
$\left(\lambda_{12}^{(A)}\right)^2$	4
$\lambda_{12}^{(1)}\,\lambda_{S_1}^{(A)}$	4
$\lambda_{12}^{(1)}\,\lambda_{S_2}^{(A)}$	4
$\left(\lambda_{12}^{(\times)}\right)^2$	4
$\lambda_{12}^{(\times)}\,\lambda_{S_1}^{(A)}$	4
$\lambda_{12}^{(\times)}\,\lambda_{S_2}^{(A)}$	4

β_{λ_5}

$\mathcal{S}$	$\mathrm{T3-E}\ (\alpha=2)$
$\lambda_5\,\lambda_1$	2
$\lambda_5\,\lambda_{HS_1}^{(1)}$	4
$\lambda_5\,\lambda_{HS_2}^{(1)}$	4
$\lambda_5\,\lambda_{12}^{(1)}$	2
$g_2^2\,\lambda_5$	$-\frac{33}{2}$
$g_1^2\,\lambda_5$	$-\frac{33}{2}$
$\lambda_5\,\lambda_{HS_1}^{(A)}$	-2
$\lambda_5\,\lambda_{HS_2}^{(A)}$	2
$\lambda_5\,\lambda_{12}^{(A)}$	2
$\lambda_5\,\lambda_{12}^{(\times)}$	-2
$\lambda_5\,T$	2

Additional scalar-coupling beta functions

$\beta_{\lambda_{HS_1}^{(A)}}$

$\mathcal{S}$	$\mathrm{T3-E}\ (\alpha=2)$
$g_1^2 g_2^2$	12

$\beta_{\lambda_{HS_2}^{(A)}}$

$\mathcal{S}$	$\mathrm{T3-E}\ (\alpha=2)$
$g_1^2 g_2^2$	24

$\beta_{\lambda_{12}^{(A)}}$

$\mathcal{S}$	$\mathrm{T3-E}\ (\alpha=2)$
$g_1^2 g_2^2$	48

$\beta_{\lambda_{12}^{(\times)}}$

$\mathcal{S}$	$\mathrm{T3-E}\ (\alpha=2)$
g_2^4	6
$\left(\lambda_{12}^{(A)}\right)^2$	2
$\lambda_{12}^{(1)}\,\lambda_{12}^{(\times)}$	8
$\left(\lambda_{12}^{(\times)}\right)^2$	10
$\lambda_{12}^{(\times)}\,\lambda_{S_1}^{(A)}$	-2
$\lambda_{12}^{(\times)}\,\lambda_{S_2}^{(A)}$	-2

Full Weinberg-coefficient RGE in EFT1

Writing the canonical mixed LLSS coefficient as C_{12}, the coupled one-loop Weinberg equation used in the F-first EFT is

$$16\pi^2\frac{dC_5}{d\ln\mu} = (2\lambda_1 - 3g_2^2 + 2T)C_5 \tag{1}$$

$$-\frac{3}{2}\left[(Y_e^\dagger Y_e)^T C_5 + C_5(Y_e^\dagger Y_e)\right] \tag{2}$$

$$+ R_W\,\lambda_5\,C_{12}. \tag{3}$$

Note: $\frac{7}{2}$ thing isnt there if we integrate out our fermion F so its not here

Direct $C_{12}\to C_5$ recoupling factor

For each representation the source term is

$$16\pi^2\,\beta_{C_5} \supset R_W\,\lambda_5\,C_{12}.$$

$\mathcal{S}$	$\mathrm{T3-E}\ (\alpha=2)$
R_W	$-\frac{4\sqrt{6}}{3}$

.

$\mathrm{T3-E}\ (\alpha=2)$

2 independent complex matched structures displayed.

complex matched operator	matched prefactor	$SU(2)_L$ invariant
$(L_i^T C L_j)\,S_1^\dagger\,S_2 + \mathrm{h.c.}$	$\frac{\sqrt{3}y_1 y_2}{3M_F}$	ϵ
$(L_i^T C L_j)\,S_1^\dagger\,S_2 + \mathrm{h.c.}$	$\frac{\sqrt{6}y_1 y_2}{3M_F}$	$\mathcal{I}_{LLS_1S_2}$

Next threshold

Further matching depends on the next field or fields selected by the threshold plan. Once all T3 fields are removed, their representation dependence is carried by the matched C_5 boundary condition and the subsequent SMEFT running is universal.